

29.

Ans: C

$$p = mv$$

$$\frac{\Delta p}{p} = \frac{\Delta m}{m} + \frac{\Delta v}{v}$$

$$\frac{\Delta p}{p} = \frac{\Delta v}{v} \quad (\because \Delta m = 0)$$

$$\Delta p \Delta x \gtrsim h$$

$$\left(\frac{9.11 \times 10^{-31} v}{100} \right) \left(\frac{0.20}{1000} \right) \gtrsim 6.63 \times 10^{-34}$$

$$v = 364 \text{ m s}^{-1}$$

30.

Ans: C